This way, we never do an HTTP call to our server (and in turn to the DB) to check if a contact is on our address book, and only read from the DB once on startup (or the first time during a session that we write an email).

4.4 Concrete Data Structures

So far with the theory, but of course implementing associative arrays to be used on real systems is a completely different thing.

In theory, if the domain (the set of possible keys) is small enough, we can still use arrays by defining a total ordering on the keys, and using their position in the order as the index for a real array. For instance, if our domain is made of the words {“a”, “terrible”, “choice”}, we could sort keys lexicographically and then we would store the values in a plain string array: for instance, {“article”, “noun”, “adjective”}. If we need to represent a dictionary that only contains a value for key “choice”, we could do that by setting the values corresponding to missing keys to `null`: {`null`, “noun”, `null`}.

Usually, however, this is not the case, and the set of possible values for the keys is large enough to make it unpractical using an array with an element for any possible key’s value: it would just require too much memory, which would moreover remain largely unused.